

Portable Ice Cube Maker

Stage-wise Project Review (7th Semester)

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Aim of the Project

- To design and develop a **small and portable ice cube maker** using **thermoelectric (Peltier) cooling**.
- To produce ice cubes **without compressors or refrigerants**.
- To apply core **ECE concepts** such as:
 - Embedded systems
 - Sensors and temperature control
 - Power electronics
 - Thermal management
- To develop a system that is **safe, eco-friendly, energy-efficient**, and suitable for **small-scale and outdoor use**.

Motivation – Background & Need

- Conventional ice makers and refrigerators:
 - Are bulky and non-portable
 - Use refrigerants harmful to the environment
 - Consume high power
 - Have complex mechanical parts
- Increasing demand for **portable cooling solutions** in:
 - Outdoor travel and camping
 - Small shops and street vendors
 - Medical and emergency applications
- Motivation to explore **solid-state cooling** as an alternative.

Motivation – Technical & Academic Interest

- Advantages of thermoelectric cooling:
 - No moving parts → low maintenance
 - Compact size → easy integration
 - Silent and safe operation
- Academic benefits of the project:
 - Integration of multiple ECE subjects
 - Hands-on experience with hardware
 - Conversion of theory into a practical system
- Helps bridge the gap between **theory and real-world application**.

Current Project Status – 7th Semester

- Literature review completed on:
 - Peltier modules
 - Small-scale cooling and ice-making systems
- Finalized system concept and working principle
- Selected major components:
 - Peltier module
 - Temperature sensor
 - Heat sink and cooling fan
 - Microcontroller
- Prepared:
 - Block diagram
 - Basic thermal calculations
 - Initial circuit design approach

Future Work – 8th Semester Plan

- **Hardware Development**
 - Assembly of Peltier module with ice mold
 - Integration of heat sink and cooling fan
- **Power and Control**
 - Design and testing of Peltier driver circuit
 - Closed-loop temperature control
- **Software Development**
 - Embedded code for sensing and control
 - Safety and protection features
- **Testing and Optimization**
 - Ice formation time analysis
 - Power consumption measurement
 - Performance improvement
- Final prototype demonstration and documentation

Thank You